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|RESEARCH ARTICLE

Determinants of Food Consumption Patterns in Rural Households of Ikwerre Local Government Area, Rivers State, Nigeria

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|Abstract

This study examined the factors influencing food consumption patterns among rural households in Ikwerre Local Government Area, Rivers State, Nigeria. The objectives were to analyze the socio-economic characteristics of respondents, assess their knowledge of adequate dietary intake, identify their food consumption patterns, identify factors influencing food consumption, and determine the constraints to proper nutrition. Primary data were collected using structured questionnaires administered to 120 household heads, selected through a multi-stage sampling procedure. Descriptive statistics (frequency, percentage, and mean ratings) were used for data analysis, alongside inferential statistics including regression analysis and Pearson product-moment correlation at a 5% level of significance. Findings revealed that respondents had a mean age of 50 years and an average of 5 years of farming experience, with only 25% reporting farming as their primary occupation. Participants demonstrated relatively high knowledge of adequate dietary intake (mean = 3.19). Beverages (mean = 2.96), fats and oils (mean = 2.46) were the most frequently consumed food items. Major factors influencing food consumption included cost of food (mean = 3.65), seasonal availability (mean = 3.36), and time available for meal preparation (mean = 3.51). Key constraints to adequate nutrition were high cost of quality food (mean = 3.64) and limited income (mean = 3.52). Regression analysis indicated that socio-economic characteristics explained 47% (R^2 = 0.47) of the variation in food consumption patterns, with no individual factor reaching statistical significance ($p > 0.05$). The correlation between dietary knowledge and actual consumption patterns was weak and not statistically significant ($r = 0.153$, $p = 0.095$). These findings underscore the complex interplay between economic, social, and structural factors in shaping rural dietary habits. The study recommends enhanced nutrition education, economic policies to reduce food costs, improved access to nutritious foods, and integrated, adaptive policy frameworks to improve rural nutrition outcomes.

Keywords: Rural households |Dietary intake|Food consumption patterns |Nutrition |Rivers State

1|Introduction

Food consumption patterns are determined by the intricate interrelationship of socioeconomic, environmental, and cultural influences (Mulenga, 2025). These patterns, in turn, influence the health, productivity, and economic stability of rural households, which predominantly rely on agriculture for their livelihoods. A rural household typically refers to a family unit living in non-urbanized areas, often engaged in agricultural activities. These households possess restricted opportunities to infrastructure (Mastura et al., 2023), lower income levels (Adesiyan & Kehinde, 2024), and a reliance on locally sourced food. The determinants of food consumption patterns in rural households include a wide range of factors. Economic determinants, such as income and food prices, play a critical role in determining what foods are accessible. Households with higher incomes are likely to have a more diverse diet, whereas low-income families may be restricted to basic staples. Social and cultural determinants, including traditional food preferences and family structure, influence the types of food consumed (Ikudayisi, Okoruwa, & Omonona, 2019). Cultural beliefs and practices often guide meal composition, food preparation, and consumption, preserving indigenous diets rich in grains, tubers, and legumes.

Environmental determinants, such as seasonal variations and agricultural productivity, affect food availability. Seasonal variations and fluctuations in agricultural productivity affect the availability of certain foods. In rural areas like Ikwerre LGA, where the economy is largely agrarian, the availability of food is closely tied to the success of farming cycles, which are impacted by weather conditions, soil fertility, and other environmental variables (Ikudayisi, Okoruwa, & Omonona, 2019). Poor harvests or climate-related disruptions can lead to reduced food availability and force households to modify their diets based on what is locally accessible. Additionally, urbanization has emerged as a significant determinant, altering traditional food consumption patterns.

Traditional diets in Nigeria, characterized by reliance on locally sourced grains, vegetables, and fruits, are typically nutrient-rich and support balanced diets. However, the shift toward more modern, processed foods in some areas has raised concerns about dietary diversity and nutrition adequacy (Oliveira et al., 2020). This change raises the possibility of increased nutritional deficiency, including both undernutrition and excess weight, as well as an increase in non-communicable diseases associated with diet which include diabetes, obesity and cardiovascular disease (Ecker et al., 2020; De Bruin et al., 2021; Mulenga, 2025).

In rural households, changes in food consumption patterns are likely driven by socio-economic constraints, including limited income and fluctuating food prices, which force households to prioritize affordability over dietary diversity (Chiaka, Zhen, & Xiao, 2022). These socio-economic challenges also limit the ability of rural households to access nutrient-rich foods that are often more expensive. Consequently, many households continue to rely on staple foods like cassava, yams, and maize, which provide basic caloric needs but may lack essential vitamins and nutrients (Onuegbu & Ibeabuchi, 2021).

Traditional food in Nigeria, reported in the study by Oladiran & Emmambux (2022) are characterized by heavy reliance on locally made, minimally processed food products such as grains, tubers, vegetables, fruits, and legumes. These food products contain a significant amount of essential nutrients, vitamins, and dietary fiber, contributing to balanced and diverse diets that support optimal health and well-being (Onuegbu, & Ibeabuchi, 2021). However, as rural households adopt more urbanized dietary patterns, it may contribute to the decline in production and consumption of locally grown and indigenous food.

It is of critical importance to understand the dynamics of food consumption in the rural areas in order to have a healthy rural population on who 70% of our food production in Nigeria rely on. This study aims to address this gap by examining the factors that influence food consumption patterns among rural households in Ikwerre Local Government Area of Rivers State.

The broad objective of the study was to determine the factors influencing food consumption among rural households in Ikwerre Local Government Area. The specific objectives of this study are therefore to Identify the socioeconomic characteristics of the respondents in the study area, to ascertain the respondent's knowledge on adequate dietary intake, to ascertain the respondent's frequency of food consumption, to identify factors influencing decision in food consumption and finally to identify constraints to adequate dietary intake among the respondents.

This study tested the following hypotheses at 0.05 significance level

$H0_1$: There is no significant relationship between socioeconomic characteristics and the food consumption patterns of rural households in the study area.

$H0_2$: There is no significant relationship between respondents' knowledge of adequate dietary intake and the food consumption pattern in the study area

2| Methodology

The study area encompasses Rivers State, one of Nigeria's 36 states. Situated between latitudes 4°15'N and 5°45'N and longitudes 5°22'E and 7°35'E, Rivers State is bordered to the South by the Atlantic Ocean, to the north by Imo and Abia States, to the East by Akwa Ibom State, and to the West by Bayelsa and Delta States. Rivers State, part of the Niger Delta, is characterized by flat plains and an intricate network of rivers and tributaries. Its tropical climate, abundant waterways, and fertile land, residents maintain strong agricultural traditions, especially in fishing, farming, commerce, and industry (Rivers State, 2023). The study focused on heads of rural households, specifically families with farmland in the Ikwerre Local Government Area (LGA) of Rivers State. A multi-stage sampling procedure was used to select participants from the study area. In stage one, Ikwerre LGA was purposively selected from the 23 LGAs in Rivers State due to the rapid urbanization in the area. The second stage used random proportionate sampling to select 10 households chosen from each of the 12 communities within Ikwerre LGA. A total of 120 respondents was therefore selected for the study. This study focuses on rural households led by individuals who own farmland in the communities of Ikwerre LGA, Rivers State, Nigeria. Data collection was done with structured questionnaire, and for illiterate respondents, an interview schedule was utilized. Data were analyzed using descriptive statistics and mean ratings, while regression analysis explored the relationship between socio economic characteristics, knowledge of dietary intake and food consumption patterns. Participants provided informed consent prior to involvement, ensuring they understood the study's aims and benefits. They were assured of the confidentiality and anonymity of their identities and responses.

3| Results and discussion

3.1| Socioeconomic Characteristics of the Respondents

The socio-economic characteristics of the rural household heads (Table 1) in this study offer insights into the demographic and economic profiles of the population, which can be compared with findings from similar studies conducted in recent years. The balanced gender distribution, with equal representation of males and females, reflects a gender parity that is consistent with other studies in similar rural settings. Oladimeji et al. (2015), observed a similar gender balance in their study of rural households, suggesting that both men and women are equally involved in household decision-making and economic activities in such communities.

The age distribution highlights that the majority of respondents are in the older age brackets,

Table 1. Socio-economic Characteristics of Respondents

Socio-economic Characteristics	Frequency (n=120)	Percentage (%)	Mean
Gender			
Male	60	50.0	
Female	60	50.0	
Age			
20-35 years	14	11.67	
36-45	17	14.17	**50 years**
46-55	32	26.67	
56-65	34	28.33	
66-75	22	18.33	
76-85	1	0.83	
Marital status			
Single	35	29.17	
Married	65	54.17	
Divorced	9	7.50	
Widowed	11	9.17	
Educational Qualification			
Tertiary	63	52.5	
Secondary	48	40.0	
Primary	5	4.2	
Non-formal	4	3.3	
Occupation			
Government job	15	12.5	
Private job	33	27.5	
Artisan/trade	25	20.8	
Farming	30	25.0	
Others	17	14.2	
Farming Experience			
0-3 years	43	35.8	
4-6	43	35.8	**5 Years**
7-10	27	22.5	
above 10 years	7	5.8	
Household Size			
1-4	19	15.83	
5-8	27	22.50	**5 Persons**
9-12	42	35.00	
13-16	32	26.67	
Income (₦)			
<50,000	65	54.17	
51,000-100,000	32	26.67	**₦36,354.17**
101,000-150,000	12	10.00	
>150,000	11	9.17	
Expenditure (₦)			
10000-40000	63	52.50	
41000-80000	34	28.33	**₦25,408.33**
81000-120000	17	14.17	
121000-160000	6	5.00	
Membership of Cooperative			
No	56	46.7	
Yes	64	53.3	

specifically between 46-65 years, with a notable 28.33% in the 56-65 age range. This indicates that the average age of respondents is 50 years, suggesting that the population may have significant life experience, which can influence their perspectives and decisions, particularly in the context of agricultural practices and economic activities. This aligns with the findings by Egwuonwu & Akwiwu (2024), who had similar population findings in their research on agricultural households.

Educational attainment among the rural household heads is relatively high, with over half having attained tertiary education. This is consistent with the findings of Okorie and Akwiwu (2025), who noted an increasing trend of higher educational attainment in rural and semi-urban areas. The high level of education in this study is likely to impact respondents' food consumption patterns, as more educated individuals may be more aware of nutrition and health, thus influencing their dietary choices. The occupational distribution in Ikwerre LGA, with significant numbers engaged in private jobs and agriculture-based occupations, aligns with the occupational patterns documented by Okorie et al. (2024) in their study of rural economies. The diversity of occupations and the relatively young workforce with moderate experience levels suggest that urbanization may be driving shifts in employment and income-generating activities, which in turn affects food consumption patterns.

Household sizes are generally large, with 35% of respondents having between 9-12 members, leading to an average household size of 5 people. This suggests that many respondents are responsible for larger family units, which can influence their economic decisions and resource allocation in agricultural practices (Chiaka, et al., 2022). Large household sizes may influence how income is spent on food, potentially leading to changes in consumption patterns as urbanization progresses in Ikwerre LGA.

Income levels show that over half of the respondents (54.17%) earn less than ₦50,000, with an average income of ₦36,354.17. This low-income distribution may indicate financial constraints that could impact their capacity for agricultural investment and improvements in their livelihoods. Correspondingly, expenditure patterns reveal that most respondents (52.5%) spend between ₦10,000-₦40,000 monthly, with an average expenditure of ₦25,408.33. Such financial dynamics can significantly affect the sustainability and growth of agricultural activities within this community. This is similar to income distributions reported by Okorie and Akwiwu (2025), who studied income and expenditure patterns in Nigerian households. The variations in income levels could contribute to differences in food consumption patterns, as lower-income households limit access to diverse food options.

3.2|Knowledge on Adequate Dietary Intake among Rural Households in Ikwerre LGA

The findings (Table 2) from this study reveal that respondents with a mean score of 3.47 know that carbohydrates are a primary source of energy for the body. This is consistent with findings from Chiaka et al. (2022), who also reported high awareness of carbohydrate functions among rural populations in Nigeria. Additionally, the awareness that carbohydrates can raise blood sugar levels, with a mean score of 3.31, suggests that the respondents are aware of the health implications of too much carbohydrate consumption, a concern that aligns with the findings of Ola et al. (2021) in their study on rural dietary patterns.

Table 2. Mean Distribution of Rural Household heads according to their Knowledge of Adequate Food Intake

Knowledge of Adequate Food Intake	SA	A	D	SD	Sum	Mean
Carbohydrates provide energy for the body	68	41	10	1	416.00	3.47
Too much carbohydrates can raise blood sugar level	57	45	16	2	397.00	3.31
Proteins are essential for body growth and repair	49	58	5	5	413.00	3.44
For both adults and children's proteins (such as beans) give	46	60	10	4	388.00	3.23
Lack of carbohydrates and protein in your food can cause you to have weak muscles and low energy	60	46	12	2	404.00	3.36
Extra energy is stored in the body as fat	40	54	22	4	370.00	3.08
fat and oil (such as palm oil, groundnut oil, egusi seed are energy sources	41	54	19	6	399.00	3.32
Excess oily fatty food can contribute to weight gain and heart diseases	61	48	8	3	407.00	3.39
Fruits and vegetables are sources of vitamins and mineral which protects our body from health issues	60	51	6	3	408.00	3.40
Including vitamins in my diet through fruits and vegetables helps the body fight against illness /diseases	58	52	7	3	404.00	3.39
Minerals like calcium are found in dairy products and vegetables, which is essential for muscle function, strong bones and teeth	55	47	14		393.00	3.28
Minerals are found in vegetables (spinach and beans), gives iron which helps carry oxygen in the blood, and for keeping your energy level up	44	57	14	4	383.00	3.22
milk and other dairy products are important for bone health	60	51	7	2	409.00	3.41
Food that are rich in fibre (such as okra, vegetables, etc) are good for the body	56	55	5	4	403.00	3.36
	82	33	2	3	434.00	3.62
Water assists in removing waste products from the body	94	23	1	2	449.00	3.74
Water helps to keep the body balanced and running smoothly					433.00	3.61
Water carries nutrients to the body cells	82	31	5	2	416.00	3.47
Grand Mean						**3.19**

Decision Mean < 2.5 =Low knowledge, $\geq$ 2.5 High Knowledge

Regarding protein intake, the respondents showed substantial knowledge about its importance for body growth and repair, with a mean score of 3.44. This mirrors the results of Okojie et al. (2017), who found that rural households had a good understanding of the role of proteins in nutrition. The slightly lower awareness of proteins' role in both adult and child nutrition, with a mean score of 3.23, indicates a potential area for further education, though it remains in line with observations from Uloh et al. (2023), who highlighted gaps in protein-related knowledge in some rural communities. Knowledge of fats and oils, such as those from palm oil and groundnut oil, being energy sources was fairly well understood, with a mean score of 3.32. This finding is consistent with that of Dedehouanou and McPeak (2020), who reported similar levels of awareness in rural areas regarding the energy-providing functions of fats and oils. The respondents' knowledge of the protective role of fruits and vegetables in preventing health issues, with a mean score of 3.40, aligns with the findings of Onyeji and Sanusi (2020), who observed high levels of awareness and knowledge about the nutritional benefits of fruits and vegetables especially in preventing disease such as cancer, heart disease and obesity. The data also revealed a strong understanding of the importance of fiber-rich foods, such as okra, in maintaining health, with a mean score of 3.36, and the highest score observed was for the knowledge that food items high in fiber are essential for health, with a mean score of 3.62. This indicates a particularly strong awareness in this area, which is in line with the findings of Sani & Scholz (2022), who also reported high knowledge of fiber's health benefits among rural households. The strong awareness of water's role in bodily functions, also reflected in the mean scores of 3.61 and 3.47 for water helping to keep the body balanced and carrying nutrients to cells, respectively, underscores the importance placed on hydration and its health benefits in rural areas, as similarly reported by Mekonnen et al. (2021).

3.3|Frequency of Food Consumption among Rural Households in Ikwerre LGA

Overall, the findings reveal that respondents' diets are diverse, with a notable preference for fats, oils, beverages, and vegetables (Table 3). The relatively lower consumption of root tubers, grains, and dairy products may be influenced by cultural, economic, or accessibility factors. Regarding diet quality, there was a higher consumption of beverages, drinks, and condiments, followed by fats and oils, vegetables and fruits, and meats, fish, and their products.

Table 3. Mean Distribution of Food Consumption among Rural Household Heads

Food Groups	Mean Score
Root Tubers	1.99
Grains & Cereals	1.96
Legumes, Pulse, and Nuts	2.29
Meats, Fish, and their Products	2.37
Milk and Dairy Products	1.69
Vegetables and Fruits	2.40
Fats & Oil	2.46
Sugar/Confectioneries/Paste	2.08
Beverages/Drinks/Condiments	2.96

These figures suggest a trend towards a relatively high intake of processed foods and fats, which are commonly found in beverages and condiments, as well as in oils. The increased consumption of such items may be indicative of a diet that relies heavily on processed or convenience foods, which can have health implications such as increased risks of obesity, cardiovascular disease, and other diet-related conditions. These findings are consistent with the findings of Nwagbo et al., (2023) who found that majority of Nigerians in peri-urban areas consumed more of beverages, drinks, and condiments, due to convenience. This aligns with findings from Petrikova et al. (2023), who reported a high consumption of sugar in Nigerian households, particularly in rural areas where it is used in various local beverages and foods. The frequent consumption of sugar reflects its role as a common sweetener and energy source in many diets, especially in tea and other beverages. However, the high intake of sugar has raised concerns about potential health implications, including increased risks of diabetes and obesity, as highlighted by Madlala et al. (2022), who emphasized the need for public health interventions to reduce sugar consumption in Africa.

This trend is supported by the research of Petrikova et al. (2023), who found that beef and fish are integral to Nigerian diets due to their availability and cultural acceptance. The preference for fish, particularly in coastal regions, is further emphasized by the findings of Fıçıcılar (2024) and Naeem & Selamoglu (2023), who noted that fish is often more accessible and affordable than other meats, contributing to its high consumption levels. Vegetables, particularly pepper and onions, also had high consumption frequencies, with mean scores of 3.70 and 3.40, respectively. This is consistent with the findings of Mekonnen et al. (2021), who reported that vegetables like pepper and onions are essential components of Nigerian cuisine, used in almost all local dishes. The high consumption of these vegetables underscores their importance in adding flavor and nutritional value to meals (Mekonnen et al., 2021).

3.4 | Factors Influencing Decision in Food Consumption among Rural Households in Ikwerre LGA

The results underscore the significant role of economic and environmental factors in influencing food intake among respondents (Table 4). The cost or price of food products emerged as the highest factor influencing decision on food consumption, with a mean score of 3.65. This finding aligns with studies such as that of Nwagbo et al., (2023), who reported that rising food prices significantly impact dietary choices, particularly in low-income households. The importance of time, seasonal changes, and food availability, reflected by a mean score of 3.36, is also notable, supporting the work of Asare-Nuamah (2021), who found that seasonal variability and food accessibility are critical determinants of food security in rural areas.

Climate change, with a mean score of 3.23, further highlights the environmental challenges affecting food consumption, consistent with the findings of Lefe et al. (2024), who linked climate variability to reduced agricultural productivity and food availability. The availability of processed and fast foods (mean score of 3.13) and traditional food preferences (mean score of 3.11) indicate a blend of modern and traditional influences on dietary habits, which mirrors the trends observed by Ikudayisi, & Okoruwa (2021) in their study on urban and rural dietary patterns in Nigeria.

Although knowledge and awareness of nutritional value, exposure to urban media and lifestyle, and access to food markets are recognized as factors influencing food intake, their impact appears to be less pronounced compared to economic and environmental factors. This is in line with research by Adekunmi et al. (2021), which highlighted that while nutritional awareness is growing, economic constraints often limit the ability to translate this knowledge into healthier food choices.

Table 4. Mean Distribution of Rural Household heads according to Factors Influencing change in Food intake

Factors	SA	A	D	SD	Sum	Mean
Knowledge and awareness of nutritional value of food	37	49	28	6	357.00	2.98
Time, seasonal change and availability of food	50	64	5	1	403.00	3.36
Climate change	46	56	18		388.00	3.23
The cost/price of food products	81	36	3		438.00	3.65
Access to food markets	31	69	19	1	370.00	3.08
Availability of processed and fast foods	41	54	25		376.00	3.13
Lack of adequate food regulation	39	68	12	1	385.00	3.21
Traditional food preference	34	69	13	4	373.00	3.11
Exposure to urban media and lifestyle	35	53	26	6	357.00	2.98
Access to information education	24	67	27	2	353.00	2.94
Personal values and beliefs	33	75	10	2	384.00	3.20
Feeding habit	42	69	9		397.00	3.31

Decision Mean < 2.5 = Not a Factor, ≥2.5 A Factor

3.5 | Constraints to Adequate Dietary Intake among the Rural Households in Ikwerre LGA

Table 5 reveals that economic factors are the most significant constraints to proper dietary intake among respondents. The incessant rise in food prices and the high cost of quality food, with a mean score of 3.64, is the most critical barrier. This finding is consistent with Ikudayisi & Okoruwa (2021) and Adekunle et al. (2020), who found that increasing food prices significantly hinder access to a healthy diet in Nigeria. Additionally, income inequality, with a mean score of 3.52, exacerbates the issue by restricting access to nutritious food for lower-income individuals. This observation aligns with that of Adekunle et al. (2020), who highlighted that economic disparities contribute to food insecurity and impact nutritional intake. Time constraints, indicated by a mean score of 3.51, also pose a significant challenge. This suggests that busy schedules and limited time for meal preparation prevent respondents from maintaining a proper diet. Mekonnen et al. (2021) support this, noting that time limitations often lead to reliance on less healthy, convenient food options.

Limited access to fresh produce, with a mean score of 3.37, further complicates dietary choices. Fadare et al. (2019) emphasize that the restricted availability of fresh fruits and vegetables affects dietary quality, as certain areas face difficulties accessing these healthy food options. Other notable constraints include disability or health conditions (mean score of 3.38), cultural beliefs and practices (mean score of 3.33), and food insecurity (mean score of 3.16). These factors also influence dietary habits, as detailed by Mekonnen et al. (2021), who found that physical health issues and cultural practices significantly impact food choices and access.

| H0[1]: There is no significant relationship between socioeconomic characteristics and the food consumption patterns of rural households in the study area

The regression analysis in Table 6 examines the relationship between various socio-economic characteristics and the consumption pattern of the respondents. The R-square value of 0.47 suggests that approximately 47% of the variation in consumption patterns can be explained by the socio-economic factors considered in the model. This indicates a moderate level of explanatory

Table 5. Mean Distribution of Rural Household heads according to the Constraints to Adequate Dietary Intake

Constraints	Strongly Agreed	Agreed	Disagreed	Strongly Disagreed	Mean
Lack nutritional education and counselling	46	41	27	6	3.06
Income inequality	74	38	4	4	3.52
Incessant rise in price and high cost of quality food	84	32	1	3	3.64
Limited access to fresh produce	42	49	18	11	3.37
Food insecurity	37	65	18		3.16
Cultural beliefs and practices	33	69	11	7	3.33
Time constraints for meal preparations	67	47	6		3.51
Disability/ health condition	64	43	8	5	3.38
Limited access healthy food retailers	28	60	17	15	2.84
Inadequate food fortification and supplementation	33	66	16	5	3.06
High level of post-harvest loss due to storage and transportation	35	61	18	6	3.04
Over reliance on staple foods like rice, cassava, and flour products	43	64	11	2	3.23

Decision Mean < 2.5 = Not a Constraint, ≥ 2.5 A Constraint

power. The F-value of 0.514, with the corresponding degrees of freedom, suggests that the overall model is not statistically significant, meaning that the combined effect of all the socio-economic variables on consumption patterns is weak. The regression reveals that all the socio-economic characteristic has p-values greater than 0.05, indicating that individually, these factors do not have a significant impact on the consumption patterns of the respondents.

Similar findings are reported in studies examining the relationship between socio-economic factors and consumption patterns. For example, a study by Ahmed & Abah (2014) found that socio-economic characteristics such as income and education have only modest effects on dietary patterns, with variability in consumption patterns attributed more to cultural and environmental factors than to socio-economic attributes (Ahmed & Abah, 2014). Similarly, Apeh et al. (2023) observed that while socio-economic variables like income and education are relevant, their impact on food consumption patterns is often moderated by other factors such as food preferences and availability (Apeh et al., 2023). In contrast, research by Purushotham et al. (2022) showed that specific socio-economic characteristics, such as income level and household size, had more pronounced effects on food consumption patterns, highlighting the importance of these variables in influencing dietary choices.

Table 6. Regression result of the relationship between the Socio-economic Characteristics of and Consumption Pattern

		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
(Constant)		2.153	0.195	-	11.025	0.000
X_1	Age	-0.034	0.047	0.099	-0.725	0.470
X_2	Marital Status	-0.017	0.051	-0.039	0.322	0.748
X_3	Educational Level	0.028	0.050	0.058	0.569	0.571
X_4	Occupation	-0.001	0.030	-0.003	-0.031	0.975
X_5	Experience	0.039	0.024	0.074	1.618	0.109
X_6	Expenditure	-0.031	0.062	-0.068	0.494	0.623
X_7	Monthly Income	0.058	0.053	0.162	1.317	0.191
X_8	Household size	-0-.077	0.063	-0.166	-1.314	0.192
	R^2	**0.47**				
	Df	**114**				
	F-value	**0.514**				
	N	**120**				

Significant at 0.05.

Table 7. Correlation result of the relationship between Respondents' Knowledge of Adequate Dietary Intake and Consumption Pattern

		Knowledge of proper dietary Intake	Consumption Pattern
Knowledge of proper dietary Intake	Pearson Correlation	1	0.153
	Sig. (2-tailed)		0.095
	N	120	120
Consumption Pattern	Pearson Correlation	0.153	1
	Sig. (2-tailed)	0.095	
	N	120	120

Significant at 0.05

|H0₂: There is no significant relationship between respondents' knowledge of adequate dietary intake and the food consumption pattern in the study area

The Table presents the correlation results between respondents' knowledge of proper dietary intake and their consumption patterns. The Pearson correlation coefficient between these two variables is 0.153, indicating a weak positive relationship. This suggests that as knowledge of proper dietary intake slightly increases, there is a minimal increase in the likelihood of better consumption patterns. However, the significance level for this correlation is 0.095, which is above the conventional threshold of 0.05 for statistical significance. This means that the observed correlation is not statistically significant, implying that the relationship between knowledge of dietary intake and consumption patterns could be due to chance. While there is a small positive association between knowledge and consumption patterns, it is not statistically significant at 0.05.

Apeh et al. (2023) found similar weak correlations between dietary knowledge and actual eating behaviors, suggesting that knowledge alone does not strongly influence consumption patterns. Similarly, a study by Akwiwu et al. (2023) observed that while there is some positive correlation between dietary knowledge and healthier eating habits, the relationship was weak and not statistically significant in their sample (Asare et al., 2021). In contrast, a study by Apeh et al. (2023) found stronger correlations between dietary knowledge and consumption patterns, suggesting that increased awareness can lead to more significant changes in eating behavior when accompanied by targeted interventions and support. However, these variations highlight the complexity of the relationship between dietary knowledge and actual consumption patterns, indicating that knowledge alone may not be sufficient to effect substantial changes without additional factors.

4 | Conclusion

The findings underscore the significant impact of urbanization on dietary habits in rural areas, highlighting a gradual shift from traditional food sources to more processed and urbanized food options. This transition is intricately linked to broader socio-economic factors, including income disparities, the rising cost of food, and time constraints. Despite the rural setting, the influence of urbanization is evident, with rural households increasingly adopting consumption patterns that mirror urban trends, often at the expense of nutritional quality. The study also revealed that, although the respondents have a baseline knowledge of proper dietary intake, this knowledge does not significantly influence their food consumption decisions. This disconnect suggests that socio-economic challenges and urban influences overshadow nutritional awareness in shaping dietary behaviours. The implications for rural and agricultural development are profound, as they highlight the need for integrated approaches that address both economic and educational deficits while maintaining the nutritional integrity of rural diets. In essence, the study highlights the critical intersection of urbanization, economic constraints, and food consumption patterns in rural development, calling for targeted interventions that support sustainable agricultural practices and enhance food security in rural communities.

| Recommendations

i. Enhanced nutritional education programs should be implemented to enable rural households to acquire adequate dietary knowledge and practice in dietary choices.

ii. Implement economic policies that reduce the cost of quality food and address income inequality to improve access to nutritious diets in rural areas.

iii. Improve infrastructure and market access to ensure that rural households have better access to fresh produce, thereby supporting healthier food consumption patterns.

|Acknowledgments

Not applicable.

|Conflict of interest

The authors declare that they have no conflict of interest.

|Data availability statement

The data used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

|Ethical approval

The University of Port Harcourt Research Ethics Committee, after critical review of this research proposal approved that it should be carried-out.

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